Cyber Physical Robotics

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Cyber Physical Robotics

Today:

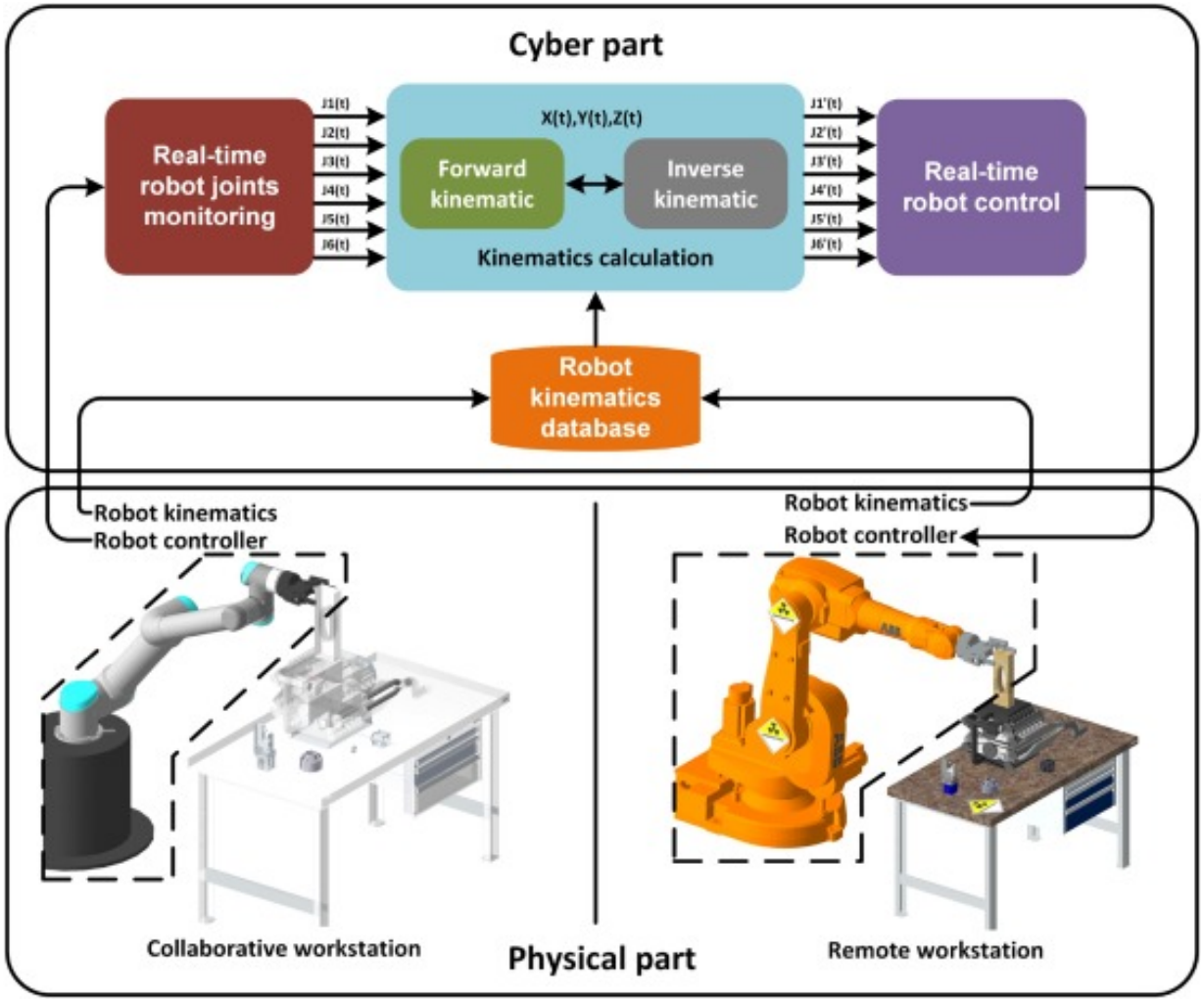
- Course organization
- Introduction to CPR

Course Objectives

- Understand what constitutes Cyber Physical Robotics (CPR)
- Grasp the mathematics and formalisms of CPRs
- Analyze and design CPR systems

Cyber Physical Robotics

- Cyber-Physical Systems made up of Robots

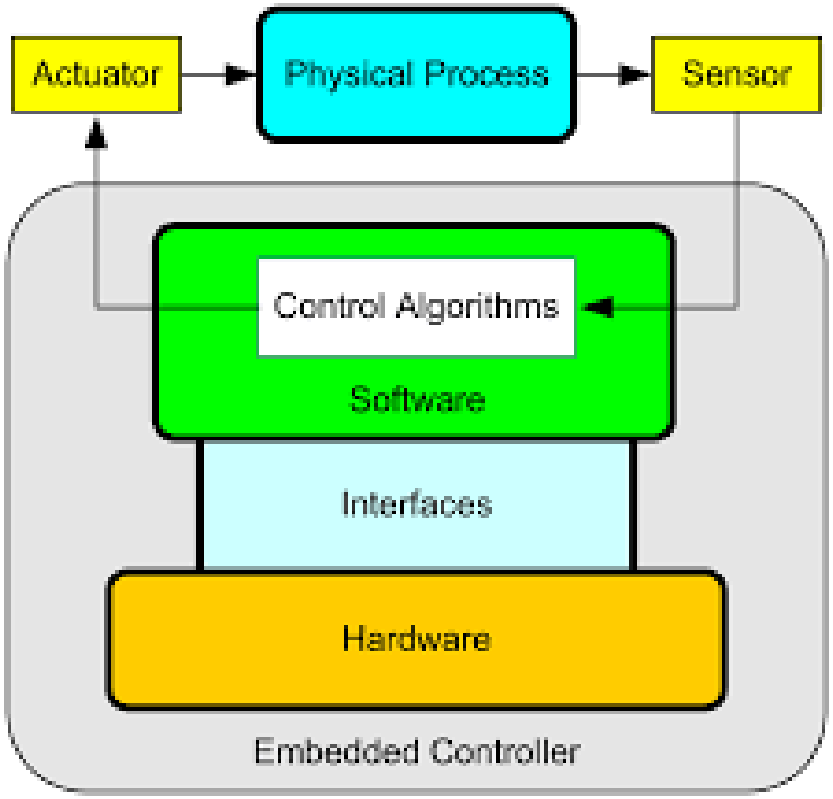


Cyber Physical Robotics

- What are Cyber-Physical Systems?

Cyber Physical System

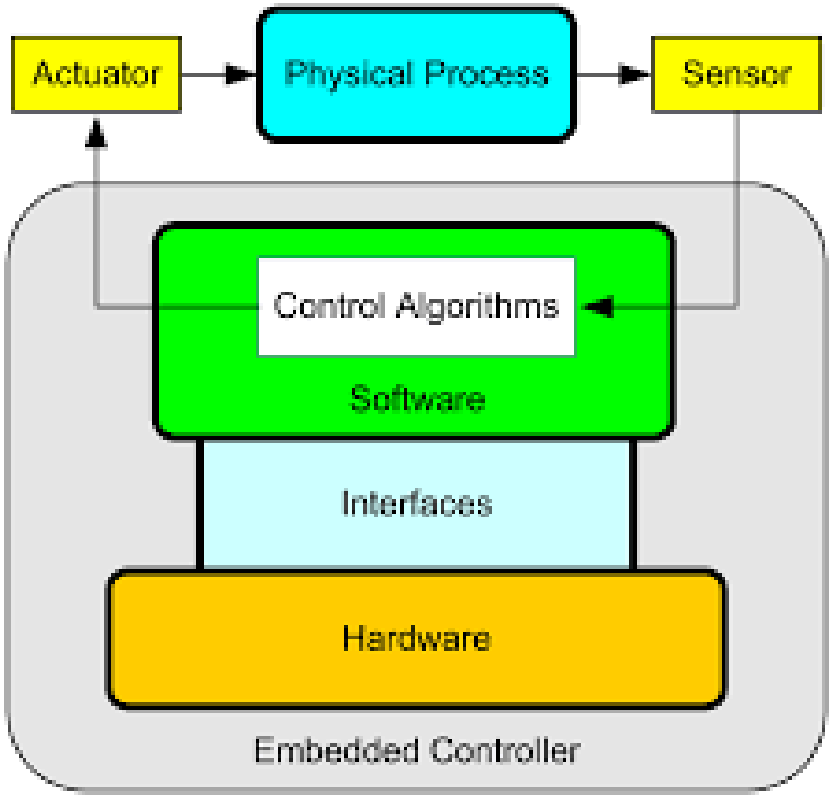
- Local control loop: actuators + sensors + controller + physical system



Cyber Physical System

Same as:

- Local control loop: embedded system + physical system = **process** (on an **embedded system**)



Cyber Physical System

Processes can communicate:



Cyber Physical Robotics

- Having concurrent processes that communicate...
- ...is not enough to make a Cyber Physical System

Cyber Physical Robotics

Cyber Physical Systems

- Arise when the controlled physical environment is **shared**

Cyber Physical Robotics

Cyber Physical System

- Concurrent processes operating on a shared physical environment

Cyber Physical Robotics

Cyber Physical System

- Concurrent processes operating on a shared physical environment
  - Emergent properties are of particular interest
    - Synchronization
    - Deadlock analysis
    - Stochastic versus deterministic properties

Cyber Physical Robotics

Cyber Physical Robotics

- Concurrent robots operating on a shared physical environment
  - Emergent properties are of particular interest
    - Synchronization
    - Deadlock analysis
    - Stochastic versus deterministic properties

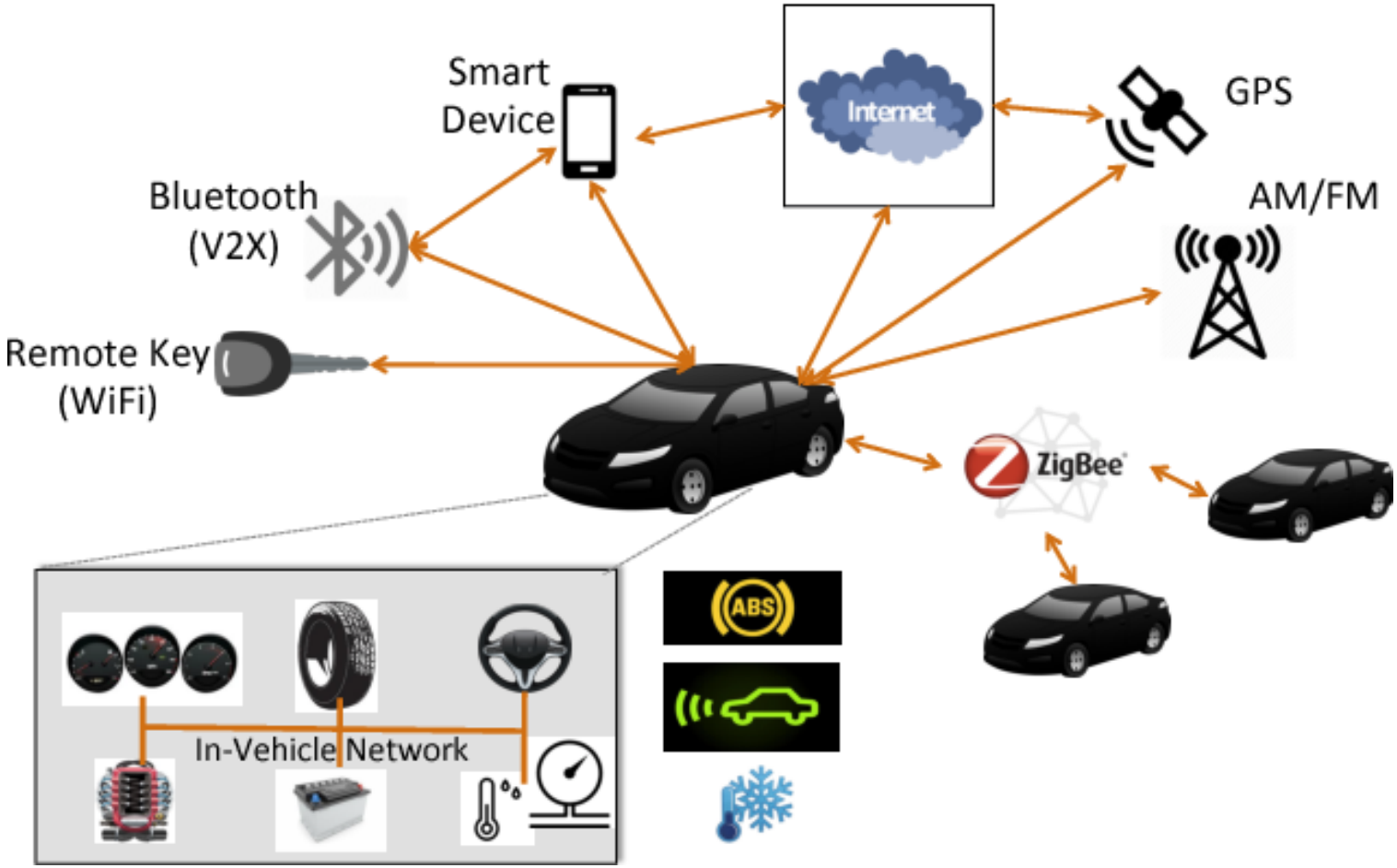
Cyber Physical Robotics

Example: industrial automation



Cyber Physical Robotics

Example: autonomous vehicles



Cyber Physical Robotics

Objectives of this course?

Cyber Physical Robotics

- Physical processes that must be controlled and observed
- (Robotic) sensors and actuators
- Local embedded systems for control
- Communication network across the distributed systems
- Global computation and control

Cyber Physical Robotics

- Physical processes that must be controlled and observed
  - Physics for Engineers, Statics, Dynamics
- (Robotic) sensors and actuators
- Local embedded systems for control
- Communication network across the distributed systems
- Global computation and control

Cyber Physical Robotics

- Physical processes that must be controlled and observed
- (Robotic) sensors and actuators
  - Industrial Robots, Sensors and Actuators for Robotic and Automation
- Local embedded systems for control
- Communication network across the distributed systems
- Global computation and control

Cyber Physical Robotics

- Physical processes that must be controlled and observed
- (Robotic) sensors and actuators
- Local embedded systems for control
  - Design of Microprocessor-Based Mechanical Systems, Modern Control and Digital Control Systems
- Communication network across the distributed systems
- Global computation and control

Cyber Physical Robotics

- Physical processes that must be controlled and observed
- (Robotic) sensors and actuators
- Local embedded systems for control
- Communication network across the distributed systems
  - Internet of Things
- Global computation and control

Cyber Physical Robotics

- Physical processes that must be controlled and observed
- (Robotic) sensors and actuators
- Local embedded systems for control
- Communication network across the distributed systems
- Global computation and control
  - We'll focus on this

Cyber Physical Robotics

Specifically, this course is about the "big picture" (i.e., emergent properties)

- Distributed System design
- Analysis and simulation
- Global computation and control

Cyber Physical Robotics

Think of this course as "Computer Science for Roboticists"

- Theory of computer science...
  - Multi-process scheduling
  - Concurrent and distributed systems
  - Process calculi
  - etc.
- ... applied to groups of robots

Cyber Physical Robotics

The tools for the job:

- Discrete Mathematics
- Probability and Statistics
- Software to implement them

Course organization

This is a project-based course

- No formally scheduled exams
- Project is individual
- 3 submission checkpoints
  - 4 weeks in
  - After midterms week
  - By last week of classes

Course organization

Project:

- A set of simulations
  - Language/tools completely up to you
- Report on the mathematics that drive those simulations, explanation

Course organization

Project description:

- See attached document in MCV

Course organization

Project presentations

- I'd like to train you in presentations
- We'll see how this goes

Next week

- Communication and network technologies